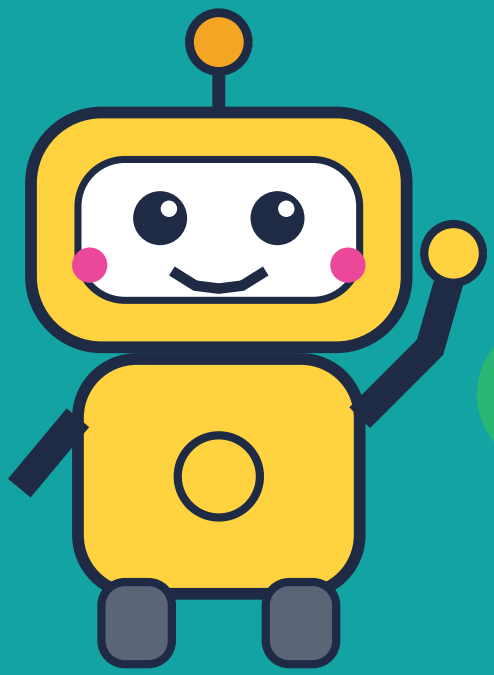
SPECIMEN BOOKLET

Think Like a Coder

Computational thinking for Classes 3–5

A sample for schools: three full chapters, a teacher script, the classroom kit and how assessment works



For CBSE's Computational Thinking and AI curriculum, 2026-27

Follows the NCERT Maths Mela chapters · No computers needed

Practice Pack Rs 449 per child (indicative) · Free pilot, January-March 2027

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CodeSchool is a working name. Not an official CBSE or NCERT publication.

A complete CT&AI programme for Classes 3-5

From 2026-27, CBSE asks schools to teach Computational Thinking and AI in Classes 3-8: **50 hours a year in Classes 3-5**, built into Mathematics and The World Around Us periods. CBSE's free handbooks give puzzles for each maths chapter. CodeSchool adds everything a teacher needs to teach it well, with no computers needed.

What schools need	CBSE's free handbook	CodeSchool Practice Pack
Puzzles for each maths chapter	Yes	Yes, in levels from easy to challenge
Theory in simple words	Short introduction	A Learn page and Robo's tip in every chapter
Worked examples	No	One per chapter, step by step
Coding	No (follow steps only)	'Now you build it': coding on paper with code cards
Hands-on activities	One or two a year	A classroom kit used all year
Teacher lesson plans	Answer keys	A period-by-period teacher script book
Assessment and report card	No format given	Self-check, observation journal and term summary
Teacher training	CBSE workshops	School orientation, termly workshops and support

★ Why schools choose it

- ✓ Follows the NCERT *Maths Mela* chapters in the same order, so it fits the maths period.
- ✓ Built around the CT skills in CBSE's curriculum: steps, patterns, abstraction, decomposition, logic and data.
- ✓ Works in any classroom: paper, cards and a floor mat. A projector helps but is optional.
- ✓ Ready for a maths or EVS teacher, not only a computer specialist.

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What each school receives

For	What they get	Classes
Each child	Practice Pack: level workbook with coding cut-outs and take-home challenges	3, 4, 5
Each class section	Classroom kit: code cards, floor grid mat, cipher wheels, sorting cards, box	3, 4, 5
Each teacher	Teacher script book, answer keys, observation journal, short how-to videos	3, 4, 5
The school	Orientation at school, a workshop every term, a support line and a term summary format	All

The series

Book	Levels	Coder skills
Class 3 Practice Pack (this sample)	B1-B3	Steps and codes, views, loops, decomposition, patterns, events, data, logic, sorting, if-then, variables, grids
Class 4 Practice Pack	B3-B5	If-else choices, nested loops, sorting machines, variables and inputs
Class 5 Practice Pack	B5-B7	Variables, search strategies, debugging and a game designed on paper
Classes 6-8 Projects and AI Pack	C1-C8	Functions, data, the AI chapters and two 20-hour projects a year (separate sample)

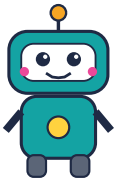
Price and ordering

Indicative price: Rs 449 per child per year (MRP) for the Classes 3-5 Practice Pack, through the school's book list. It can replace an existing computer book, so the cost to parents stays about the same. Kits, teacher books and training come with the packs. Prices are indicative until confirmed in writing.

Try it first

A **free pilot** in two or three sections from January to March 2027: materials, a teacher orientation, weekly support and a short report against success measures agreed in advance. No obligation to continue.

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Hello! I am Robo. I think like a coder, and soon you will too! In this book we will solve puzzles, crack secret codes, guide robots and write programs, all on paper.

How this book works

Learn — Robo explains one big idea in simple words, with the new coder words to remember.

Example — A worked example shows the idea step by step.

Try it! — Five practice questions. Circle the answer or write it in the box.

Now you build it — Coding on paper: put code cards in order, fix bugs and write your own programs.

Challenge — A harder puzzle for when you are ready for more.

How did I do? — Colour a face to show how sure you feel.

My code cards

Coders use cards like these. Each colour means something different. You will find cut-out cards at the back of the book.

Step: do this | **Repeat: a loop** | **If: a decision** | **When: an event** | **move**

For teachers and parents

Each chapter goes with the same chapter of the NCERT Class 3 maths book, *Maths Mela*, so it can be used in maths periods. CBSE suggests about 50 hours of computational thinking a year in Classes 3-5. The coding here needs no computer: children act out, write and check programs on paper, which builds the same ideas used later in block coding and AI. Answers are at the back. Encourage children to explain their thinking, not just the answer.

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How a lesson runs

Every lesson fits one 40-minute period and follows the same five steps, based on the PRIMM approach used in computing classrooms (predict, run, investigate, modify, make). Children learn the routine once and then focus on the thinking.

1. **Predict and run** 8 min
   Children predict what a set of steps will do, then act it out.

2. **Investigate** 8 min
   Pairs answer 'why' questions about how it worked.

3. **Modify** 8 min
   Children change the steps to do something new.

4. **Put in order** 8 min
   Jumbled code cards go back in the right order.

5. **Make** 8 min
   Pairs write their own program and swap with another pair.

How it supports CBSE's assessment

CBSE asks schools to assess CT&AI through written puzzles, group activities and a **Teacher Observation Journal**. Each step gives the teacher something to observe: predictions show reasoning, modifications show understanding, and the 'make' step shows whether a child can build an algorithm of their own. The journal format and a term summary for the report card come with the teacher book.

Every child can take part

- ✓ Children who finish early move to the challenge level; children who need help use a hint card.
- ✓ Most activities are done in pairs, so children who find reading hard can still take part by talking and acting out.
- ✓ Teacher preparation is about 5-10 minutes per lesson using the script.

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Inside the full Class 3 book

The complete book has 14 chapters, one for each NCERT *Maths Mela* chapter, plus answers, a glossary and cut-out code cards. This sample includes chapters 1, 11, 14 in full.

Ch	Chapter	Coder skill	Level	In this sample
1	What's in a Name?	Algorithms and secret codes	B1	Yes
2	Toy Joy	Abstraction: seeing what matters	B1	
3	Double Century	Patterns and loops	B1	
4	Vacation with My Nani Maa	Breaking problems into steps	B1	
5	Fun with Shapes	Shape patterns and repeat	B2	
6	House of Hundreds - I	Events: when this happens, do that	B2	
7	Raksha Bandhan	Equal groups and sorting data	B2	
8	Fair Share	Rules for sorting and sharing	B2	
9	House of Hundreds - II	Clues and logical thinking	B2	
10	Fun at Class Party!	Sorting: putting things in order	B2	
11	Filling and Lifting	If-then rules and comparing	B3	Yes
12	Give and Take	Variables: keeping count	B2	
13	Time Goes On	Sequences and loops in time	B2	
14	The Surajkund Fair	Grids and robot routes	B1-B2	Yes

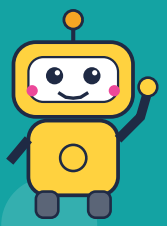
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CHAPTER 1 · LEVEL B1

1 What's in a Name?

Coder skill: Algorithms and secret codes

Goes with NCERT Maths Mela, Class 3, chapter 1

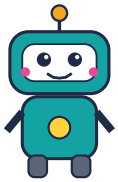


Learn

When you brush your teeth, you follow steps in the same order every day. A list of steps like this is called an **algorithm**.

A **secret code** also follows an algorithm. In a **shift code**, every letter moves forward in the alphabet by the same number. That number is the **key**. Only someone who knows the key can read the message.

algorithm	code	key
steps done in order	a hidden message	the secret number



An algorithm is just a list of steps. Follow them in order and you get the same answer every time!

Example

Code the name RAM with key +1

1. Take the first letter, **R**. Move one letter forward: **S**.
2. Next letter **A** becomes **B**. Then **M** becomes **N**.
3. So **RAM** is coded as **SBN**. To read it back, move each letter one step *back*.

top row: plain letters bottom row: coded (+1)

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A

★ *Coder fact!* Julius Caesar used a shift code more than 2,000 years ago. He moved each letter 3 places forward!

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Try it!

Circle the right answer, or write it in the box.

1 Use key **+1** to code the word **BAT**.

2 Mira coded a word with key **+2**. Her code is **DQQM**. What is the word?

3 Robo coded **SUN** as **TVO**. What key did Robo use?

a) +1 **b)** +2 **c)** +3 **d)** +4

4 Write these names from **shortest to longest**: Meera, Ali, Joseph, Tara.

5 Look at the first letters: **A, C, E, G, ?** Which letter comes next?

a) H **b)** I **c)** J **d)** K

Now you build it

These steps code any word with key +2, but they are jumbled. Number them in the right order.

- Write the new letter down
- Repeat until no letters are left
- Take the next letter of the word
- Move it 2 letters forward

Now use your steps to code your own name with key +2.

Challenge

Robo coded **HELLO** as **KHOOR**. What was the key? Use the same key to decode **FDW**.

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✓ How did I do?

I can follow steps in order.	I can code and decode a word.	I can find the key of a code.
		

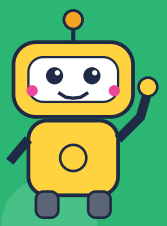
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CHAPTER 11 · LEVEL B3

11 Filling and Lifting

Coder skill: If-then rules and comparing

Goes with NCERT Maths Mela, Class 3, chapter 11

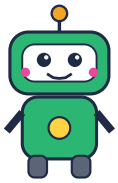


Learn

An **if-then** rule makes a decision: *if* something is true, *then* do this. If it is raining, then take an umbrella.

A **balance** helps us compare weights. The heavier side goes **down**. If both sides are level, they weigh the same.

if-then	heavier	capacity
a rule that makes a decision	weighs more	how much a thing can hold

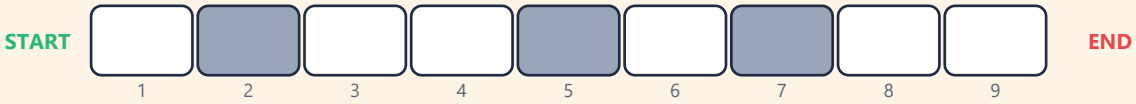


If the jug is full, then I stop pouring. If-then rules help me decide what to do next!

Example

The tile rule

1. Rule: *if you are on a white tile, then hop 1. If you are on a grey tile, then hop 2.*
2. Start on tile 1 (white): hop to 2. Tile 2 is grey: hop to 4. Tile 4 is white: hop to 5.
3. Each tile tells Robo what to do next.



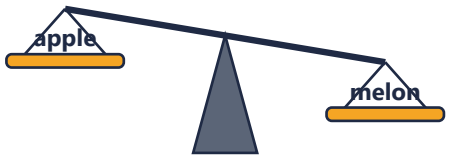
★ *Coder fact!* Washing machines use if-then rules: if the drum is full of water, then stop filling.

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Try it!

Circle the right answer, or write it in the box.

1 Which is heavier?



a) apple **b)** melon **c)** same

2 Use the tile rule from the example on this strip. Start on tile 1. Which tiles does Robo land on?



3 Rule: *if it is sunny, then wear a cap, else take an umbrella*. What do you take on a rainy day?

a) cap **b)** umbrella **c)** both **d)** nothing

4 A bucket fills 8 mugs. A jug fills 3 mugs. Which holds more?

a) bucket **b)** jug **c)** same

5 The bag is heavier than the book. The book is heavier than the pencil. Which is the lightest?

a) bag **b)** book **c)** pencil

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Now you build it

Write if-then rules for watering a plant.

If the soil is dry | then pour one mug | If the soil is wet | then ___

What should happen if the pot is already full of water?

Challenge

You have 4 bags and a balance. What is the smallest number of weighings to be sure which bag is the heaviest?

How did I do?

I can follow an if-then rule.	I can compare weights on a balance.	I can compare how much things hold.
		

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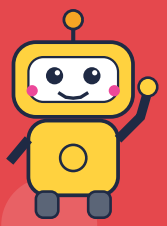
14

CHAPTER 14 · LEVEL B1-B2

The Surajkund Fair

Coder skill: Grids and robot routes

Goes with NCERT Maths Mela, Class 3, chapter 14

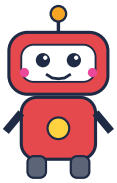


Learn

A **grid** has rows and columns. We can name a square with a letter and a number, like **C2**.

A **program** is a set of instructions for a computer or a robot. Robo moves one square for each arrow. A program that hits a wall has a bug!

grid	program	route
rows and columns of squares	instructions a robot follows	the path you take

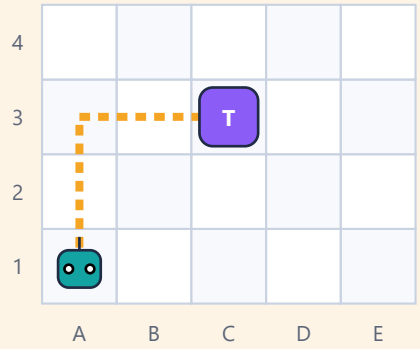


Give me arrow cards and I will follow them exactly. One arrow = one square. That is my program!

Example

Take Robo to the toy stall

1. Robo is at **A1**. The toy stall (T) is at **C3**.
2. Program: up, up, right, right.
3. Robo moves 4 squares and reaches the stall. Another route works too: right, right, up, up.



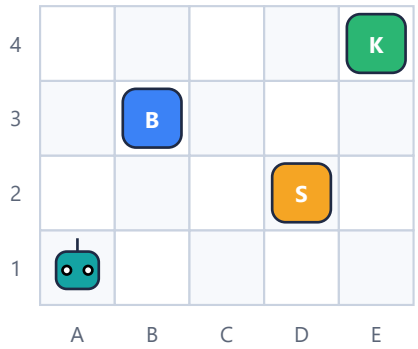
★ *Coder fact!* Robots in big warehouses move around grids, following programs, to fetch parcels.

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Try it!

Circle the right answer, or write it in the box.

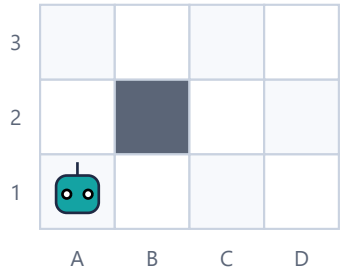
1 Robo starts at A1 and follows: **right, right, up, right**. Which stall does Robo reach?



a) S (sweets) **b)** B (bangles) **c)** K (kites)

2 Write the **shortest** program to take Robo from A1 to the kite stall K at E4 in the grid above.

3 Robo is at A1 and the grey square is a wall. Program: **up, right, right**. Where is the bug?



a) the 1st arrow **b)** the 2nd arrow **c)** the 3rd arrow **d)** no bug

4 Make this program shorter with a loop: **right, right, right, right**.

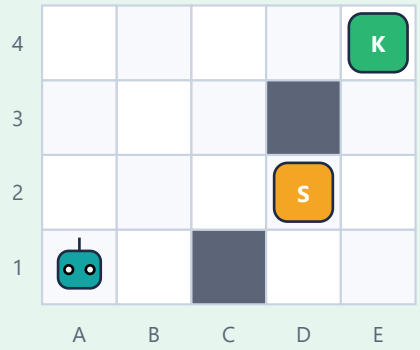
5 Which square is 2 to the right of B1 and 1 up?

a) D2 **b)** C2 **c)** D1 **d)** B3

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Now you build it

Plan Robo's trip around the fair. Start at A1, visit the sweet stall S, then the kite stall K, without crossing the grey walls.



up down left right

Draw your arrows in order. Count them. Can a friend find a shorter route?

Challenge

Write a program with a loop that takes Robo from A1 to D4 along the diagonal staircase: right, up, right, up, right, up.

How did I do?

I can name squares on a grid.	I can follow and write a robot program.	I can find a bug in a route.
☺ 😐 ☹	☺ 😐 ☹	☺ 😐 ☹

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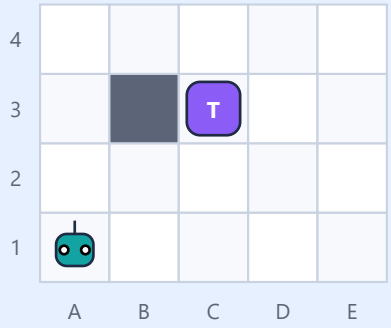
Sample teacher script: Robot routes

From the teacher book, Class 3, chapter 14 (*The Surajkund Fair*). One 40-minute period. Level B1-B2.

You need	
Kit	Floor grid mat (or a 4 x 5 grid drawn on the floor or board), arrow cards, two stall cards
Book	Chapter 14 pages, 'Try it' questions 1-3
Goal	Children follow and write a program of arrows, and find a bug in a route

Step and time	Teacher does	Children do
1. Predict and run (8 min)	Put 'Robo' (a child) on A1 and a stall card on C3. Hold up: up, up, right, right.	Predict where Robo will stop. One child walks the program while the class reads the arrows aloud.
2. Investigate (8 min)	Ask: 'Would right, right, up, up reach the same stall? Does the order matter?'	Discuss in pairs. Try it on the mat. (Here both work; with a wall, order can matter.)
3. Modify (8 min)	Put a 'wall' card on B3. Ask children to change one arrow so Robo avoids it.	Change the program and test it on the mat.
4. Put in order (8 min)	Give each pair jumbled arrow cards for a new stall.	Order the cards so Robo reaches the stall, then do 'Try it' question 1.
5. Make (8 min)	Ask pairs to plan a trip to two stalls.	Write the route with arrow cards, swap with another pair and test their route.

Lesson set-up



Set-up: Robo starts on A1, the toy stall (T) is on C3. In step 3, put the wall card on B3 (grey). Children then change one arrow so Robo still reaches the stall.

Common mistakes to watch for

- **Counting the start square as a step.** Ask: 'Did Robo move yet?'
- **Mixing up left and right.** In this level, arrows are map directions (up means towards the top of the grid), not turns.
- **Skipping a square when walking the route.** Have the class count each move aloud.

Different paces

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Support: use a 3 x 3 grid and only up and right arrows.
Stretch: shorten a route with 'repeat' (for example, repeat 3 times: right). Ask which route is shortest and how they know.

Check for understanding

Two quick questions at the end of the period:

1. Robo is on A1. Write the arrows to reach C2. 2. With a wall on B2, the program *up, right, right* fails. Which arrow is the bug? (the 2nd)

Note in the observation journal: **Beginning** (needs help following arrows), **Developing** (follows, cannot yet write), **Secure** (writes a working route), **Extending** (finds the shortest route or uses repeat).

Curriculum link

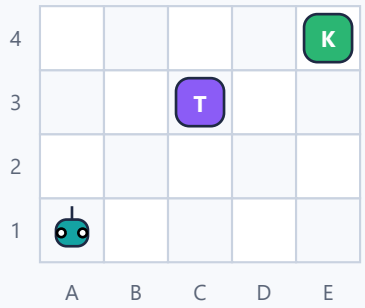
Grids and robot routes connect to CBSE's Class 3 handbook chapter on *The Surajkund Fair* (paths, grids and pathway tiles), and build the idea of a program as exact instructions: the base for block coding in later classes.

Teacher notes

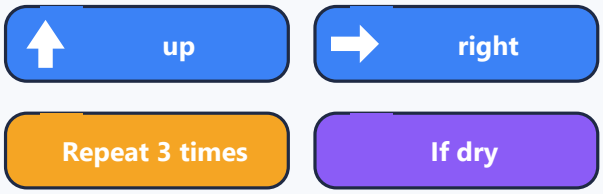
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The classroom kit

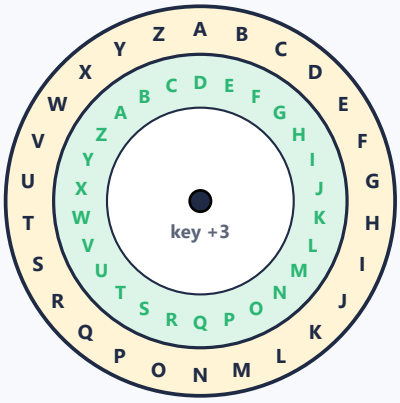
One kit per class section, used all year. Everything is printed, laminated and reusable.



Floor grid mat (6 x 6 ft) for robot routes



Code cards: arrows, repeat, if, when (4 decks)



Cipher wheels for secret codes (20)



Picture and shape cards for patterns and sorting (120)

Item	Quantity per kit
Code card decks (playing-card size, laminated)	4 decks of 54
Cut-out coding blocks (die-cut sheets)	20 sheets
Floor grid mat, washable	1
Cipher wheels	20
Picture and shape cards	120
Robot direction tokens	30
Teacher quick-guide card	1
Sturdy box	1

Illustrations show the design. Final materials will be confirmed after the pilot.

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Assessment and the report card

CBSE asks for continuous, competency-based assessment in CT&AI. The teacher book includes an observation journal using four levels, and a term summary that fits on the report card.

Skill	Beginning	Developing	Secure	Extending
Following steps	Follows short steps with help	Follows and fixes simple steps	Writes steps with loops	Designs and improves own steps
Patterns	Spots simple repeats	Continues a pattern	Finds the core and rule	Makes new patterns and explains them
Breaking problems down	Needs help to start	Breaks familiar jobs into steps	Breaks new problems into steps	Plans steps and their order alone
Logic and rules	Uses one clue	Uses two clues or an if-then rule	Uses several clues and if-else	Spots extra clues and explains

Sample term summary

Area	Level	Teacher comment
Computational thinking (overall)	Secure	Breaks problems into steps and writes loops confidently.
Coding on paper	12 of 14 'build it' tasks	Ready for if-else rules with more than two choices.
Self-check ('How did I do?')	Mostly green	Finds codes easy and clues harder; we practised clue puzzles together.

Children check themselves too

Each chapter ends with **'How did I do?'**, where children colour a face for each 'I can' statement. Teachers can compare this with what they observed, which helps children learn to judge their own progress.

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Answers for the sample chapters

1. What's in a Name?

1. CBU
2. BOOK
3. a) +1
4. Ali, Tara, Meera, Joseph
5. b) I
Build it: Take the next letter; Move it 2 letters forward; Write the new letter down; Repeat until no letters are left
Challenge: Key +3; FDW = CAT

11. Filling and Lifting

1. b) melon
2. 1, 2, 4, 6, 7 then hop 2 to the end
3. b) umbrella
4. a) bucket
5. c) pencil
Build it: Example: If the soil is wet, then do not water
Challenge: 3: compare bags 1 and 2, then the heavier with bag 3, then the heavier with bag 4

14. The Surajkund Fair

1. a) S
2. 7 arrows: 4 right and 3 up, in any order
3. b) the 2nd arrow hits the wall at B2
4. Repeat 4 times: right
5. a) D2
Build it: Example: up, right, right, right (S), right, up, up (K)
Challenge: Repeat 3 times: right, up

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Cut-out code cards

Ask an adult to help you cut along the dotted lines. Use the cards to build programs on your desk, then act them out with a friend.

up	down	left
up	down	left
right	Start	Repeat ___ times
right	Start	Repeat ___ times
If ___	else	When ___ happens
If ___	else	When ___ happens
add ___	take away ___	Stop
add ___	take away ___	Stop

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A note for parents

Schools can send this page home at the start of the year.

Dear parents,

This year your child will learn **computational thinking** as part of maths, following CBSE's new curriculum for Classes 3-8. Computational thinking means solving problems the way a programmer does: breaking a big job into steps, spotting patterns, keeping what matters and following clear rules. These skills help in maths, reading and everyday life, and they are the base for coding and AI later.

Your child will use the *Think Like a Coder* workbook in class. There are no screens: children solve puzzles, crack secret codes, guide a robot around a grid and write programs with paper code cards.

How you can help at home

- ✓ Ask your child to explain a puzzle to you. Explaining is part of the learning.
- ✓ Play 'robot' at home: your child gives you step-by-step instructions to make a sandwich, and you follow them exactly.
- ✓ Spot patterns together: on clothes, tiles, calendars and number plates.
- ✓ Praise effort and good thinking, not just right answers.

Take-home challenge

Write the numbers 1 to 20. Colour every 3rd number. What pattern do you see? Which number would be the 10th coloured one?

Answer: 3, 6, 9, ... the 10th is 30.

No child's personal data is collected by the workbook. Thank you for your support.

Class teacher _____ **School** _____

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Next steps for your school

1. **Look through this sample**
   Try a chapter with a class or ask a teacher to review it.

2. **Talk to us**
   A short meeting with the principal and coordinator to answer questions.

3. **Start a free pilot**
   Two or three sections, January to March 2027, with an orientation and weekly support.

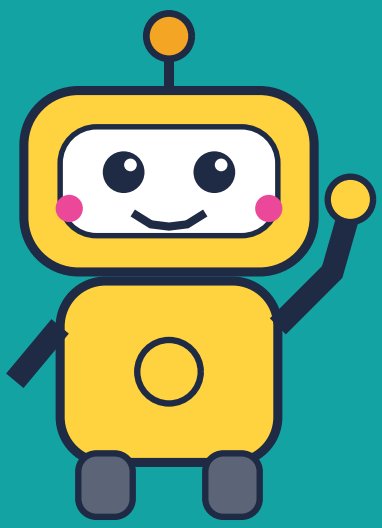
4. **Add it to the 2027-28 book list**
   Most schools finalise their list by the end of January.

Contact

Anoop Dobhal, CodeSchool, Bullseye Knowledge System Limited
Phone / WhatsApp: [Phone] · Email: [Email]

Notes

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Think Like a Coder

Computational thinking for Classes 3-5, for CBSE schools

Ask us about a free pilot

Phone / WhatsApp: [Phone] · Email: [Email]

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